SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Uppara Vinay Kumar	**Reg. No.:**	192311348
Section / Batch:	CSE (4th Year)	**Date:**	05/10/2026

Code of Conduct

I Uppara Vinay Kumar certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Individual Task: Energy-Efficient Communication in Mobile Devices and Networks

1. Introduction:

Mobile devices run on limited battery power, and cellular networks consume a huge amount of electricity to serve billions of users. Energy-efficient communication means delivering the required data rate and quality of service while using as little energy as possible, both in the mobile device and in the network.

In simple words, a phone should last longer on one charge, and the base stations and core network should waste less power when traffic is low.

2. Why Energy Efficiency Matters:

- **Battery life:** the radio interface (transmitter, receiver, and idle listening) is one of the largest power consumers in a smartphone.
- **Operating cost:** the Radio Access Network (RAN), mainly base stations, uses most of an operator's energy.
- **Environment:** lower energy use reduces carbon emissions and supports green communication.
- **Growing demand:** more users, more data, and more IoT devices increase total power consumption.

3. Basic Architecture:

Energy saving can be applied at every level of the cellular system:

Mobile Device ↔ Base Station (RAN) ↔ Core Network ↔ Internet

Device-side techniques extend battery life, while network-side techniques reduce the power drawn by base stations and core equipment.

4. Energy-Efficient Techniques in Mobile Devices:

- **Discontinuous Reception (DRX):** the phone switches its receiver off for short periods and wakes only to check for incoming data. Connected-mode DRX (C-DRX) is used in 4G LTE and 5G NR.
- **Idle and sleep modes:** when there is no traffic, the device moves to RRC Idle (or RRC Inactive in 5G) and consumes very little power.
- **Power control:** the device transmits with only the power needed to reach the base station, which saves energy and reduces interference.

- **Dynamic Voltage and Frequency Scaling (DVFS):** the processor lowers its speed and voltage when the workload is light.
- **Wi-Fi offloading and network selection:** the device uses the most energy-efficient available network for a given task.
- **Efficient application behaviour:** batching data transfers, reducing background activity, and using push notifications instead of constant polling.

5. Energy-Efficient Techniques in Networks:

- **Base station sleep mode:** lightly loaded cells or radio components are switched off at night or during low traffic and switched on when demand rises.
- **Cell zooming and load balancing:** coverage of cells is adjusted so that neighbouring cells can take over traffic from the ones put to sleep.
- **Small cells and heterogeneous networks:** short-range, low-power cells deliver data efficiently where it is needed.
- **Cloud RAN (C-RAN):** baseband processing is centralised and shared, which reduces equipment and cooling energy.
- **Efficient hardware:** high-efficiency power amplifiers and low-power components in the radio units.
- **Renewable energy:** solar and other renewable sources power remote base stations.

6. 4G and 5G Energy-Saving Features:

4G LTE

- DRX and idle mode for devices; cell on/off and load balancing in the eNodeB.
- Adaptive modulation and coding, so the transmission matches the channel quality.

5G NR

- **Beamforming and Massive MIMO:** energy is directed toward the user instead of being spread in all directions.
- **Lean carrier design:** fewer always-on reference signals so that base stations can sleep more often.
- **Bandwidth Part (BWP) adaptation:** the device uses a narrow bandwidth when traffic is low, saving power.
- **Wake-up signals and RRC Inactive state:** the device sleeps deeply and wakes quickly only when needed.

- **Network slicing and virtualisation:** resources are allocated only where they are needed, so unused functions can be scaled down.

7. Challenges:

- Balancing energy saving with latency, coverage, and quality of service.
- Higher frequency bands (mmWave) and more antennas increase hardware power use.
- Frequent switching between sleep and active states adds signalling overhead.

8. Conclusion:

To conclude, **energy-efficient communication is essential for both mobile devices and cellular networks**. Techniques such as DRX, sleep modes, power control, small cells, C-RAN, and Massive MIMO beamforming reduce power consumption while keeping the required performance. From 4G LTE to 5G, the architecture has evolved to provide **longer battery life, lower operating cost, and a greener network**.

9. Geotag Photo:

